

Full Length Article

Exploring unified cross-view hypergraph generation for multi-view semi-supervised classification

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ABSTRACT

Graph structure is widely used in the field of multi-view learning. Hypergraph which is a kind of extension of graph can capture the higher-order relationships of nodes in a better way. However, most existing hypergraph-based models are based on the assumption that hypergraph structures are readily available, which is untenable in most cases. In order to alleviate this problem, we propose the learnable unified hypergraph dynamic system framework, a novel approach in unified cross-view hypergraph structure generation tailored for multi-view semi-supervised classification. Specifically, we introduce four strategies for unified cross-view hypergraph generation and propose a mechanism for generating learnable unified cross-view hypergraph. Furthermore, we utilize a dynamic diffusion model to dynamically learn unified hypergraph structure which can achieve better performance in multi-view semi-supervised classification tasks. Extensive experimental results on various real datasets show that the proposed method outperforms other state-of-the-art multi-view algorithms.

1. Introduction

Graph-based techniques show significant advantages in multi-view learning (Chen, Wu, Chen, Dong, & Wang, 2023; Wang, Huang, et al., 2024; Zhao, Fei, Zhang, Wen, & Cui, 2025). By integrating data from multiple views, graph can help explore complex pairwise connections among data and it has been widely used in various practical applications such as node classification (Huang et al., 2024; Wu, Lin, Lin, Chen, Bai, & Wang, 2023; Yao, Liang, Liang, Li, & Cao, 2022), recommender systems (Lin et al., 2024; Zhang, Niu, Ma, & Zhang, 2025; Zou et al., 2022), and clustering tasks (Bian et al., 2025; Li, Chen, Wang, & Lai, 2024; Peng, Kang, Chen, Kang, Chen, & Cheng, 2024). It is worth noting that data structure in actual practice may go beyond pairwise connections (Feng, You, Zhang, Ji, & Gao, 2019). In multi-view scenarios, the relationships between data tend to be even more complex, resulting in traditional graph structure that limits the performance of multi-view learning to a certain extent. Hypergraph as an extension of graph structure can use its hyperedges to explore the higher-order data relationship that may be beyond the pairwise relationship, as shown in Fig. 1(a). To better model complex relationships between multi-view data, it is crucial to introduce hypergraph into multi-view learning.

Most existing hypergraph-based models usually assume hypergraph structures are always available in specific situations (Mei, Bi, Wen,

Kong, & Wu, 2024; Wu, Li, Zhang, Chen, Ng, & Long, 2024). However, the above well-constructed hypergraph structures are not readily available under multi-view scenarios. Therefore, a nature question is raised: *How can we construct hypergraph structures for multi-view semi-supervised classification?* The way on constructing hypergraphs for multi-view semi-supervised classification is a non-trivial task while there are challenges from several perspectives:

Challenge I: *How to generate an effective hypergraph under unsupervised scenarios to capture multi-view higher-order relationship?* A straightforward way to generate hypergraphs for multi-view learning is naturally generating view-specific hypergraph as shown in Fig. 1(b). However, this way fails to capture the complex higher-order relationship among views. As illustrated in Fig. 1(b), it is clear in view 1 that n_1 , n_2 , and n_3 are similar because they all belong to the Van Gogh, Bright, and Intense colors items. However, that overlooks the higher-order similarity of n_3 and n_4 which becomes apparent when considering view 2. How to inject such higher-order relationship among views into a unified cross-view hypergraph structure is still an unsolved problem.

Challenge II: *How to jointly refine the generated unified cross-view hypergraph and perform multi-view semi-supervised classification?* Due to the fact that the unified cross-view hypergraph constructed based on the static distribution of the original data is static, it may involve

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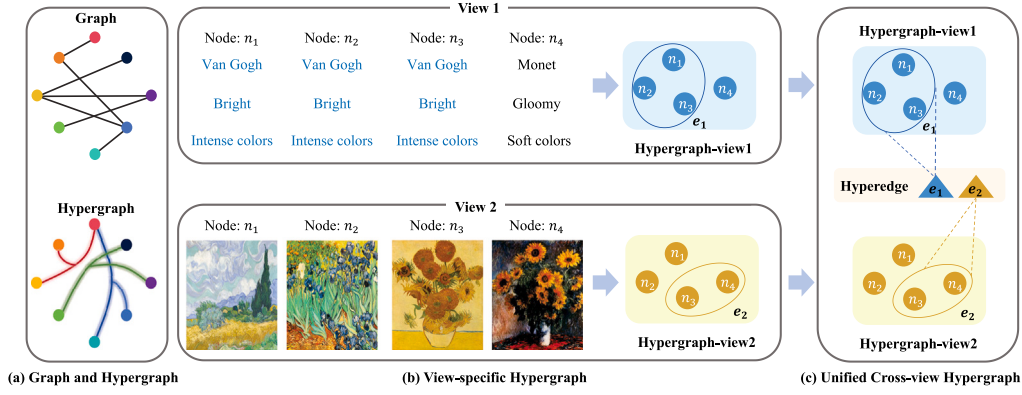


Fig. 1. An illustration of the structures of graph, hypergraph, view-specific hypergraph, and unified cross-view hypergraph.

task-irrelevant information. In the semi-supervised classification task, most existing methods construct hypergraphs in the pre-processing stage where the constructed hypergraphs are unlearnable. The static hypergraph which fails to dynamically filter task-irrelevant information may lead to inferior performance when performing multi-view semi-supervised classification. How to dynamically refine hypergraphs for semi-supervised classification to better capture task-relevant information is a problem that needs to be addressed.

In order to address the above challenges, in this paper, we devise an end-to-end framework named Learnable Unified Hypergraph Dynamic System (LUHDS) for multi-view semi-supervised classification, which consists of two pivotal components. For **Challenge I**, we introduce four strategies ($U_1 \sim U_4$) for unified cross-view hypergraph generation as shown in Fig. 1(c) and propose a mechanism (U_5) for generating a learnable unified cross-view hypergraph. For **Challenge II**, we devise a dynamic diffusion system to perform multi-view semi-supervised classification and jointly refine the generated unified cross-view hypergraph to effectively capture task-relevant information.

Our comprehensive contributions in this paper can be summarized as follows:

- We propose four ways for unified cross-view hypergraph generation and further propose a mechanism for generating learnable unified cross-view hypergraph under multi-view scenarios.
- We devise an end-to-end framework named learnable unified hypergraph dynamic system for multi-view semi-supervised classification, which can dynamically optimize the hypergraph structure to filter task-irrelevant information.
- Extensive experimental results on several widely used datasets demonstrate that the proposed method outperforms other state-of-the-art algorithms in multi-view semi-supervised classification.

The rest of this paper is organized as follows. In Section 2, we conduct a comprehensive review of multi-view learning and hypergraph neural networks. In Section 3, we elaborate on the proposed method and analyze the time complexity. In Section 4, we perform experiments on various datasets and show the analysis of experimental results. Finally, the conclusion is presented in Section 5.

2. Related work

2.1. Multi-view learning

Due to the complexity of entities in the real world, it is not sufficient to describe them from one perspective alone. Benefiting from the rapid development and widespread usage of multimedia, data can be obtained from multiple sources. Multi-view learning (Lu, Wu, Chen, Cai, & Wang, 2024; Ma, Hu, Ge, & Zhang, 2023; Zou, Fang, Wu, Zheng, & Wang, 2024) has garnered significant attention in machine learning due

to its ability to leverage diverse sources of information for improved performance.

In recent decades, there has been a notable rise in the development of multi-view learning techniques. Zhang, Qiang, Wang, and Nie (2022) constructed affinity graphs based on an anchor-based strategy and proposed a framework to learn an optimal multi-view consensus graph by using feature and label information. Wang and Fan (2024) proposed a method for multi-view graph pooling from an information theory perspective, which simplified the original graph into a more structured and disentangled coarsened graph. Wang, Guo, Wang, and Wang (2024) combined the strengths of subspace learning and structured graph learning and proposed a novel framework for multi-view clustering. Du, Cai, Wu, Pi, and Wang (2024) considered both original and generative graphs to balance consistency and diversity in multi-view learning and proposed a universal multi-view consensus graph learning framework. Fang, Du, Chen, and Wang (2024) integrated four synergistic modules into a framework to achieve generalizability and reliability beyond the known.

Although the above mentioned works have shown excellent performance in the field of multi-view learning, they are all based on traditional graph structure. Hypergraph structures are not widely applied in the field of multi-view learning, which leaves room for improvement in the performance of multi-view learning.

2.2. Hypergraph neural networks

Hypergraphs have emerged as a powerful extension of traditional graphs, tailored to model higher-order relationships among data. In recent years, there has been a growing number of works based on hypergraph structures (Feng, Han, Ying, & Gao, 2024; Gao, Feng, Ji, & Ji, 2022). Early work by Feng et al. (2019) introduced the hypergraph neural network, designing a hyperedge convolution operation to handle the data correlation. This pioneering effort laid the foundation for applying hypergraph structures to deep learning techniques, enabling the modeling of complex dependencies among multiple entities. Yadati et al. (2019) extended traditional graph convolutional network (Kipf & Welling, 2017) to hypergraph and proposed HyperGCN which was designed for semi-supervised learning. Zhang et al. (2020) conducted label propagation on hypergraph and combined deep neural networks to optimize the feature embedding. Wu, Yan, and Ng (2022) introduced the hypergraph reconstruction error as a regularizer to learn an effective classifier and achieved significant performance. Song et al. (2024) proposed a contrastive hypergraph neural network which could learn from labeled and unlabeled data by self-supervised contrastive learning techniques. Tang, Chen, Shi, and Lyu (2024) introduced a dual hypergraph message passing network for event prediction which performed the information aggregation and representation fusion on the entity hypergraph and the relation hypergraph.

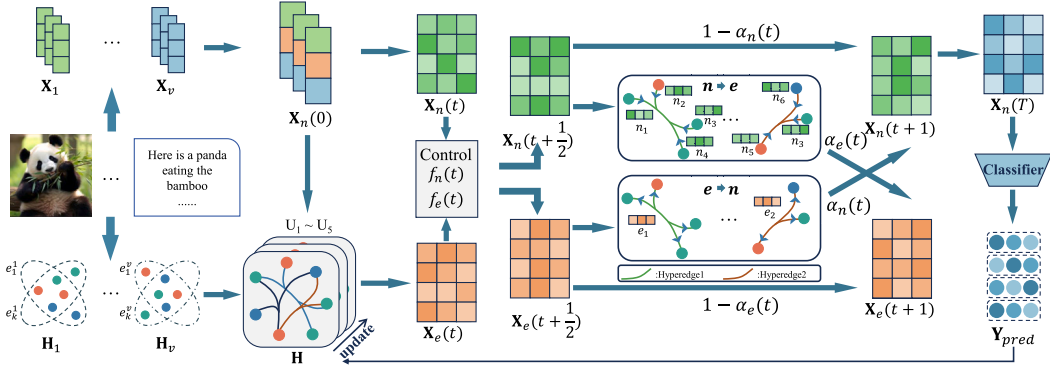


Fig. 2. The framework of the proposed method. The proposed method generates a learnable unified cross-view hypergraph, performs multi-view semi-supervised classification and jointly refines the generated unified cross-view hypergraph with dynamic diffusion system.

Table 1

A summary of primary notations in this paper.

Notations	Explanations
$\mathbf{X}_v \in \mathbb{R}^{N \times d^v}$	The given multi-view feature matrix of v -th view.
$\mathbf{X}_n(t) \in \mathbb{R}^{N \times d}$	The representation of nodes at time t .
$\mathbf{X}_e(t) \in \mathbb{R}^{N \times d}$	The representation of hyperedges at time t .
$\mathbf{H}_v \in \mathbb{R}^{N \times E}$	The unified cross-view hypergraph structure.
$\mathbf{A}(t)$	The diffusion matrix of the proposed method at time t .
$\alpha_n(t)$	The teleport probabilities of nodes at time t .
$\alpha_e(t)$	The teleport probabilities of hyperedges at time t .
$\mathbf{D}_v^{-1} \in \mathbb{R}^{N \times N}$	The average normalization matrix of node degree.
$\mathbf{D}_e^{-1} \in \mathbb{R}^{E \times E}$	The average normalization matrix of hyperedge degree.
$\mathbf{Y} \in \mathbb{R}^{N \times c}$	Real labels of the given nodes.
$\hat{\mathbf{Y}} \in \mathbb{R}^{N \times c}$	Predictive labels of nodes after learning.

The above mentioned works demonstrate the wide range of applications of hypergraph structures in various fields, illustrating the effectiveness and superiority of deep learning methods based on hypergraph. However, performing hypergraph neural networks requires hypergraph structures which are not directly given in the multi-view data. Therefore we need to consider how to construct a hypergraph structure which is suitable for multi-view data.

3. The proposed method

In this section, we elaborate on the proposed model. The main notations used in this paper are partly shown in Table 1. The proposed method aims to construct a unified cross-view hypergraph structure which leverages higher-order relationships among views and jointly performs multi-view hypergraph structure construction and semi-supervised classification. Fig. 2 depicts the overall pipeline of the proposed method.

3.1. Unified cross-view hypergraph generation

In this subsection, we introduce four ways of unified cross-view hypergraph generation and introduce how to generate a learnable unified cross-view hypergraph structure from the multi-view data. In order to explore higher-order relationship among data, how to generate an effective hypergraph structure is important. It is worth noting that the hypergraph structure is not provided directly in the multi-view data. Here, we introduce four ways to generate unified cross-view hypergraph under multi-view scenarios.

Generally, the ways of constructing hypergraph include distance-based, clustering-based, similarity-based and joint entropy based methods. In specific:

- **Distance-based Unified Cross-view Hypergraph Construction (U_1).** Hypergraph on the Euclidean distance utilizes local relationship information between nodes. Usually, k -nearest neighbor (kNN) is employed in the scenarios. Given a sample x , it can be connected by its k -nearest neighbors with a hyperedge. The unified cross-view hypergraph is defined as:

$$\mathbf{H}_u(i, e) = \begin{cases} 1, & \mathbf{x}_i^v \in kNN(\mathbf{x}_j^v), \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Here, $i = 1, 2, \dots, N$, $j = 1, 2, \dots, N$, $v = 1, 2, \dots, V$, and $e = jv$.

- **Clustering-based Unified Cross-view Hypergraph Construction (U_2).** The methods based on clustering categorize the nodes into multiple clusters and a node can be connected to nodes belonging to the same cluster through a hyperedge. Taking K-means as an example, the hypergraph is constructed as follows:

$$\min_{\mathbf{h}_k^v} \sum_{k=1}^K \sum_{\mathbf{x}^v \in \mathbf{h}_k^v} \|\mathbf{x}^v - \mu_k^v\|_2^2, \quad (2)$$

where μ_k^v is the centroid of point $\mathbf{x}^v \in \mathbf{h}_k^v$. The unified cross-view hypergraph is defined as: $\mathbf{H}_u = \{\mathbf{h}_1^1, \dots, \mathbf{h}_K^1, \dots, \mathbf{h}_1^V, \dots, \mathbf{h}_K^V\}$.

- **Similarity-based Unified Cross-view Hypergraph Construction (U_3).** There is a general consensus that nodes with similar features have the same label in most cases. Therefore nodes with similar features are better connected using hyperedges. One of the most common measures of node similarity is cosine similarity. Based on this, the unified cross-view hypergraph is defined as:

$$\begin{cases} \mathbf{H}_u(i, j) = \frac{\mathbf{x}_i^v \cdot \mathbf{x}_j^v}{|\mathbf{x}_i^v| |\mathbf{x}_j^v|}, \\ \mathbf{H}_u = \mathbf{H}_1 \parallel \mathbf{H}_2 \parallel \dots \parallel \mathbf{H}_V. \end{cases} \quad (3)$$

Here, $\parallel$ is the matrix concatenation operation.

- **Joint entropy based Unified Cross-view Hypergraph Construction (U_4).** Joint entropy captures the complex dependencies between samples and can provide a more comprehensive representation of the structure and dependencies within the data. A smaller joint entropy indicates a stronger correlation between samples. In light of this, the hypergraph is constructed as follows:

$$\mathbb{E}(\mathbf{x}_i, \mathbf{x}_j) = - \sum_{k=1}^d \sum_{\mathbf{x}_{ik} \in \mathbf{x}_i} \sum_{\mathbf{x}_{jk} \in \mathbf{x}_j} p(\mathbf{x}_{ik}, \mathbf{x}_{jk}) \log p(\mathbf{x}_{ik}, \mathbf{x}_{jk}), \quad (4)$$

$$\mathbf{H}_u(i, e) = \begin{cases} 1, & \mathbb{E}(\mathbf{x}_i^v, \mathbf{x}_j^v) < \epsilon, \\ 0, & \text{otherwise,} \end{cases} \quad (5)$$

where $\mathbb{E}(\mathbf{x}_i^v, \mathbf{x}_j^v)$ indicates the joint entropy of node i and node j in v th view, $e = jv$, and ϵ is a threshold.

Algorithm 1 The algorithm of LUHDS.

Require: Multi-view data $\mathbf{X} = \{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_V\}$, label matrix $\mathbf{Y}$, parameters k, T , the control step interval s .
Ensure: The prediction label $\mathbf{Y}_{pred} \in \mathbb{R}^{N \times 1}$.

- 1: Generate learnable unified cross-view hypergraph $\mathbf{H}_u$.
- 2: **for** $epoch = 1$ to $epochs$ **do**
- 3: Initialize $\mathbf{X}_n(0) = \mathbf{X}_1 || \mathbf{X}_2 || \dots || \mathbf{X}_V$, and $\mathbf{X}_e(0) = \mathbf{D}_e^{-1} \mathbf{H}_u^T \mathbf{X}_n(0)$.
- 4: **for** $t = 0$ to T **do**
- 5: **if** current time is initial time or time s has elapsed since the last control step **then**
- 6: Perform control step by Eq. (6).
- 7: **else**
- 8: Skip control step by:
 $\mathbf{X}_n(t + \frac{1}{2}) = \mathbf{X}_n(t)$, $\mathbf{X}_e(t + \frac{1}{2}) = \mathbf{X}_e(t)$.
- 9: **end if**
- 10: Perform diffusion step by Eq. (7).
- 11: **end for**
- 12: Calculate prediction matrix $\hat{\mathbf{Y}}$ and $\mathcal{L}$ by Eq. (9).
- 13: **end for**
- 14: **return** The predicted label of data points is given by $\mathbf{Y}_{pred} = \arg \max_j \hat{\mathbf{Y}}_{ij}$.

Learnable unified cross-view hypergraph. The above part describes four strategies to generate the unified cross-view hypergraph structure denoted as $\mathbf{U}_1, \mathbf{U}_2, \mathbf{U}_3$, and $\mathbf{U}_4$. However, the hypergraph structure is obtained based on the original static distribution of the multi-view data, which may contain task-irrelevant information leading to sub-optimal performance. Hence, we initialize a learnable unified cross-view hypergraph structure denoted as $\mathbf{U}_5$ to capture task-relevant information while performing the semi-supervised classification task.

3.2. Learnable unified hypergraph dynamic system

In order to perform multi-view semi-supervised classification based on the generated learnable unified cross-view hypergraph, we design an end-to-end framework Learnable Unified Hypergraph Dynamic System (LUHDS) inspired by the recent progress of ordinary differential equation-based dynamic diffusion model (Yan, Feng, Ying, & Gao, 2024). It can utilize higher-order relationships among views and dynamically optimize the hypergraph structure to capture task-relevant hyperedge during the training process. In the following, we present the details of the proposed method.

In the proposed method, the time iteration is separated into a control step and a diffusion step. The control step is defined as:

$$\begin{bmatrix} \mathbf{X}_n(t + \frac{1}{2}) \\ \mathbf{X}_e(t + \frac{1}{2}) \end{bmatrix} = \begin{bmatrix} \mathbf{X}_n(t) \\ \mathbf{X}_e(t) \end{bmatrix} + \begin{bmatrix} f_n(\mathbf{X}_n(t)) \\ f_e(\mathbf{X}_e(t)) \end{bmatrix}, \quad (6)$$

where $\mathbf{X}_n(t)$ and $\mathbf{X}_e(t)$ are the representations of nodes and hyperedges at time t , $f_n(\cdot)$ and $f_e(\cdot)$ is the control function of nodes and hyperedges. In the implementation, we use a single-layer multilayer perceptron (MLP) to initialize $\mathbf{H}_u \in \mathbb{R}^{N \times E}$ as $\mathbf{H}_u = MLP(\mathbf{O})$, where $\mathbf{O}$ is the Xavier-initialized embedding. And we use two single-layer fully connected networks as $f_n(\cdot)$ and $f_e(\cdot)$. Similar to residual networks, the control step alters the representations of nodes and hyperedges through control functions. In order to lower the total training duration of the method, the control step is conducted every certain number of layers. The diffusion step is defined as:

$$\begin{bmatrix} \mathbf{X}_n(t+1) \\ \mathbf{X}_e(t+1) \end{bmatrix} = \begin{bmatrix} \mathbf{X}_n(t + \frac{1}{2}) \\ \mathbf{X}_e(t + \frac{1}{2}) \end{bmatrix} + \mathbf{A}(t) \begin{bmatrix} \mathbf{X}_n(t + \frac{1}{2}) \\ \mathbf{X}_e(t + \frac{1}{2}) \end{bmatrix}. \quad (7)$$

Here, $\mathbf{A}(t)$ is a diffusion matrix. We design the diffusion matrix as:

$$\mathbf{A}(t) = \begin{bmatrix} -\alpha_n(t)\mathbf{I} & \alpha_n(t)\mathbf{D}_n^{-1}\mathbf{H}_u \\ \alpha_e(t)\mathbf{D}_e^{-1}\mathbf{H}_u^T & -\alpha_e(t)\mathbf{I} \end{bmatrix}, \text{ s.t. } 0 < \alpha_n(t), \alpha_e(t) \leq 1. \quad (8)$$

The diffusion step propagates representation messages between nodes and hyperedges based on $\mathbf{A}(t)$. The above equation can be written as $\mathbf{X}_n(t+1) = (1 - \alpha_n(t))\mathbf{X}_n(t + \frac{1}{2}) + \alpha_n(t)\mathbf{D}_n^{-1}\mathbf{H}_u\mathbf{X}_e(t + \frac{1}{2})$, $\mathbf{X}_e(t+1) = \alpha_e(t)\mathbf{D}_e^{-1}\mathbf{H}_u^T\mathbf{X}_n(t + \frac{1}{2}) + (1 - \alpha_e(t))\mathbf{X}_e(t + \frac{1}{2})$. Here, $\alpha_n(t)$ and $\alpha_e(t)$ are learnable parameters to control the teleport probabilities of nodes and hyperedges at time t , respectively. The aggregation from the hyperedge representation is denoted as $\mathbf{H}_u\mathbf{X}_e(t + \frac{1}{2})$ and the aggregation from the node representation is denoted as $\mathbf{H}_u^T\mathbf{X}_n(t + \frac{1}{2})$. The stability analysis of the diffusion step is presented in Section 3.2.1 and the optimization of $\alpha_n(t)$ and $\alpha_e(t)$ can be found in Section 3.2.2.

Once final time T of the proposed method is determined, the node representation is $\mathbf{X}_n(T)$. Then we employ a linear layer as a classifier to conduct semi-supervised classification with cross-entropy loss:

$$\begin{cases} \hat{\mathbf{Y}} = \text{classifier}(\mathbf{X}_n(T)), \\ \mathcal{L} = - \sum_{i \in \Omega} \sum_{j=1}^C Y_{ij} \ln \hat{\mathbf{Y}}_{ij}, \end{cases} \quad (9)$$

where Ω denotes the set of training data points with labels, C is the number of classes, and $\mathbf{Y}$ is the label matrix. The algorithm of the proposed method is presented in Algorithm 1.

3.2.1. Stability analysis

Due to the diffusion step reflecting the time iteration of the node representation and hyperedge representation, it is necessary to analyze the stability of the diffusion step. Here, we first analyze the proposition of the eigenvalue of the diffusion matrix $\mathbf{A}(t)$ and then perform a stability analysis on the diffusion process.

Theorem 1. Assume that the diffusion matrix's eigendecomposition is $\mathbf{A}(t) = \mathbf{U}\mathbf{A}\mathbf{U}^{-1}$ with the eigenvalue matrix $\mathbf{A} = \text{diag}(\lambda_1, \dots, \lambda_{N+E})$ and the eigenvectors $\mathbf{u}_i$ in diffusion step equation, the eigenvalue $\{\lambda_i\}_{i=1}^{N+E}$ lies in the left half-plane of the complex plane or is 0.

Proof.

$$\mathbf{A}(t) = \begin{bmatrix} -\alpha_n(t)\mathbf{I} & \alpha_n(t)\mathbf{D}_n^{-1}\mathbf{H}_u \\ \alpha_e(t)\mathbf{D}_e^{-1}\mathbf{H}_u^T & -\alpha_e(t)\mathbf{I} \end{bmatrix}, \text{ s.t. } 0 < \alpha_n(t), \alpha_e(t) \leq 1. \quad (10)$$

Consider the sum of all rows except the elements on the diagonal. For all $i \in \{1, 2, \dots, N\}$, it holds $\sum_{j \neq i} |\mathbf{A}(t)_{ij}| = \alpha_n(t)$ since $d(n) = \sum_{e \in \mathcal{E}} \mathbf{H}_u(n, e)$. Similarly, for all $i \in \{N, N+1, \dots, N+E\}$, it holds $\sum_{j \neq i} |\mathbf{A}(t)_{ij}| = \alpha_e(t)$. According to the Gershgorin circle theorem, the eigenvalue λ_i of $\mathbf{A}(t)$ satisfies $\lambda_i \in \mathbb{R}_N \cup \mathbb{R}_E$. Here

$$\mathbb{R}_N = \{z \in \mathbb{C} : |z - \mathbf{A}(t)_{ii}| \leq \sum_{j \neq i} |\mathbf{A}(t)_{ij}| \wedge i \in \{1, 2, \dots, N\}\} \quad (11)$$

$$= \{z \in \mathbb{C} : \|z + \alpha_n(t)\mathbf{I}\|_2 \leq \alpha_n(t)\},$$

$$\mathbb{R}_E = \{z \in \mathbb{C} : |z - \mathbf{A}(t)_{i,i}| \leq \sum_{j \neq i} |\mathbf{A}(t)_{ij}| \wedge i \in \{N, N+1, \dots, N+E\}\} \quad (12)$$

$$= \{z \in \mathbb{C} : \|z + \alpha_e(t)\mathbf{I}\|_2 \leq \alpha_e(t)\}.$$

Two sets $\mathbb{R}_N$ and $\mathbb{R}_E$ have similar forms, therefore they are merged further as:

$$\lambda_i \in \mathbb{R}_N \cup \mathbb{R}_E = \{z \in \mathbb{C} : \|z + \max(\alpha_n(t), \alpha_e(t))\mathbf{I}\|_2 \leq \max(\alpha_n(t), \alpha_e(t))\}. \quad (13)$$

Therefore, the eigenvalue λ_i of $\mathbf{A}(t)$ is 0 or distributed in the left half-plane of the complex plane (i.e., the real part of $\lambda_i < 0$).

Following the theory in Yan et al. (2024), the diffusion step is stable when the real part of the eigenvalue is less than 0. If the eigenvalues have additional non-zero imaginary parts, the step will oscillate and the oscillation will decrease over time. The number of 0 eigenvalues is then determined to investigate the property of the diffusion matrix.

Table 2
A brief description of the datasets used in the experiments.

Dataset	# Samples	# Views	# Classes	# View features
WebKB_Texas	187	2	5	187/1,703
GRAZ02	1,476	6	4	512/32/256/500/500/680
NUS-WIDE	1600	6	8	64/144/73/128/225/500
HW	2000	6	10	153/596/301/481/157/27
Hdigit	10,000	2	10	784/256
Esp-game	11,032	2	7	100/100

3.2.2. Optimization

In this part, we will elaborate on the parameter optimization problem on $\alpha_n(t)$ and $\alpha_e(t)$. We employ the proximal gradient descent to solve the optimization problem with constraints. We denote $g(\cdot)$ as the gradient function and calculate the gradients of $\alpha_n(t)$ and $\alpha_e(t)$, respectively. That can be written as:

$$\begin{cases} g_{\alpha_n(t)} = \nabla_{\alpha_n(t)} \mathcal{L}(\alpha_n(t), \alpha_e(t), \mathbf{X}_n(T), \mathbf{X}_e(T)) = 0, \\ g_{\alpha_e(t)} = \nabla_{\alpha_e(t)} \mathcal{L}(\alpha_n(t), \alpha_e(t), \mathbf{X}_n(T), \mathbf{X}_e(T)) = 0. \end{cases} \quad (14)$$

Assume that the optimal solution of the above equation is $\alpha_n^*(t)$ and $\alpha_e^*(t)$. In order to satisfy the feasible region $0 < \alpha_n(t), \alpha_e(t) \leq 1$, we use the proximal gradient descent. The solution of the above problem can be acquired by the following equation:

$$\begin{cases} \alpha_n(t) = \min\{\max\{\gamma_n, \alpha_n^*(t)\}, 1\}, \\ \alpha_e(t) = \min\{\max\{\gamma_e, \alpha_e^*(t)\}, 1\}. \end{cases} \quad (15)$$

Here, γ_n and γ_e are positive values that converge infinitely to zero.

3.2.3. Time complexity analysis

In this part, we will analyze the time complexity of the proposed method. The computational complexity mainly lies in matrix multiplications. The time complexity of the control step is $\mathcal{O}((|N| + |E|)d^2)$, which is mainly limited by multiplying the weight matrices and the representations. Considering $\mathbf{H}_u$ as a sparse matrix, the time complexity of the diffusion step is $\mathcal{O}((tr(\mathbf{D}_n) + tr(\mathbf{D}_e))d)$, which is limited by multiplying the hypergraph incidence matrix and the representations. When $\mathbf{H}_u$ is a dense matrix, the time complexity of the diffusion step is $\mathcal{O}((|N| + |E|)^2d)$. Here, d is the dimension of representations. Although time complexity depends on T , it can be omitted in calculations due to $T \ll N$ and $T \ll d$. Overall, the computational complexity of the proposed method costs $\mathcal{O}((tr(\mathbf{D}_n) + tr(\mathbf{D}_e))d + (|N| + |E|)d^2)$ or $\mathcal{O}((|N| + |E|)^2d + (|N| + |E|)d^2)$.

4. Experiments

In this section, we start with a brief description of datasets used in this paper and experimental settings. Then we conduct comprehensive experiments to evaluate our model. In particular, we verify the effectiveness of unified cross-view hypergraph generation in ablation study.

4.1. Datasets

In order to validate the effectiveness of the proposed method, we utilize six widely used multi-view datasets for the performance evaluation including: WebKB_Texas,² GRAZ02,³ NUS-WIDE,⁴ HW,⁵ Hdigit,⁶ and Esp-game.⁷ Details for all of these datasets are presented in Table 2.

4.2. Baselines and experimental settings

We compare the proposed LUHDS with various classical and state-of-art multi-view algorithms including MLAN (Nie, Cai, & Li, 2017), AWDR (Yang, Deng, & Nie, 2019), Co-GCN (Li, Li, & Wang, 2020), ERL-MVSC (Huang, Wang, Zheng, Zhao, & Lin, 2021), DSRL (Wang, Chen, Du, & Lin, 2022), LGCNFF (Chen, Fu, et al., 2023), and GEGCN (Lu, Wu, Zhong, et al., 2024). Among them, MLAN, AWDR, and ERL-MVSC are traditional baselines while Co-GCN, DSRL, LGCNFF, and GEGCN are networks-based algorithms.

- **MLAN**: The method employs a local structure embedding approach to construct a graph capable of performing both clustering and semi-supervised classification tasks.
- **AWDR**: This approach is designed for multi-view classification, where the features from diverse views are automatically weighted based on learned optimal values.
- **Co-GCN**: This method merges co-training and GCN, enabling adaptive combination of Laplacians to fully exploit graph information from various views.
- **ERL-MVSC**: By integrating diversity, sparsity, and consensus, this method effectively manages multi-view data while boosting robustness.
- **DSRL**: A deep learning model for sparse regularization is proposed in this method, which can adaptively learn sparse regularizers driven by data.
- **LGCN-FF**: This framework incorporates learnable GCN with feature fusion and it uses a multi-stage optimization strategy to optimize the learnable parameters.
- **GEGCN**: This framework integrates the extraction of multi-view consistency and complementarity, graph optimization, and semi-supervised classification tasks into a comprehensive framework.

All compared algorithms are tuned as their default settings. In the proposed method, the termination time T is set to 20, and the control term time interval is set to 10. The strategy of unified cross-view hypergraph generation is U_5 . Adam Optimizer is adopted to optimize parameters with learning rate $lr \in \{0.001, 0.005, 0.01\}$ and weight decay $\in \{5 \times 10^{-3}, 5 \times 10^{-4}, 5 \times 10^{-8}\}$. The range of dropout rate in our model is $\{0.05, 0.1, 0.15, 0.2\}$. For the training, we independently select 10% labeled data as the training set, 10% unlabeled data as the validation set and 80% unlabeled data as the test set, following the previous work (Wang, Huang, et al., 2024). The experiments are conducted on a system equipped NVIDIA GeForce RTX 3090 with 24 GB of VRAM and implemented by PyTorch 1.12.0.

4.3. Multi-view semi-supervised classification

In this subsection, we perform multi-view semi-supervised classification and compare the performance of the proposed method with other comparison algorithms. In order to further verify the effectiveness of the proposed method in the field of multi-view semi-supervised learning, we also conduct experiments under different supervised information scenarios.

² <https://www.cs.cmu.edu/afs/cs.cmu.edu/project/theo-11/www/wwkb/>

³ <http://www.emt.tugraz.at/pinz/data>

⁴ <https://c-trend.comp.nus.edu.sg/technologies.html>

⁵ <http://yann.lecun.com/exdb/mnist/>

⁶ <https://cs.nyu.edu/roweis/data.html>

⁷ <https://www.kaggle.com/datasets/parhamsalar/espgame>

Table 3

Classification Accuracies and F1-scores (mean% and standard deviation%) of all methods. The best results are bolded and the sub-optimal results are underlined.

Method	Metric	WebKB_Texas	GRAZ02	NUS-WIDE	HW	Hdigit	Esp-game
MLAN	ACC	56.2 (3.3)	54.2 (1.3)	<u>45.8 (1.0)</u>	92.3 (0.6)	97.1 (0.2)	70.1 (0.5)
	F1	19.5 (10.4)	54.7 (1.7)	<u>45.3 (1.6)</u>	92.5 (0.6)	97.1 (0.2)	71.2 (0.3)
AWDR	ACC	65.0 (5.5)	39.9 (2.4)	31.9 (1.2)	88.8 (1.1)	62.9 (1.6)	85.0 (0.5)
	F1	35.2 (9.2)	39.9 (2.3)	33.0 (1.1)	88.9 (1.1)	63.0 (1.5)	<u>84.9 (0.5)</u>
Co-GCN	ACC	60.5 (8.3)	48.6 (3.2)	41.7 (1.4)	85.4 (4.8)	88.8 (3.1)	75.9 (3.5)
	F1	31.4 (10.6)	47.7 (3.1)	39.7 (1.2)	85.4 (4.8)	88.8 (3.0)	75.5 (3.5)
ERL-MVSC	ACC	64.5 (4.2)	54.5 (1.5)	43.2 (1.1)	90.3 (1.1)	95.4 (0.2)	68.6 (0.4)
	F1	<u>44.7 (12.4)</u>	54.7 (1.7)	42.3 (1.3)	90.5 (1.0)	95.4 (0.2)	68.5 (0.4)
DSRL	ACC	60.0 (6.5)	48.1 (1.0)	41.5 (3.1)	95.6 (2.4)	<u>98.6 (0.1)</u>	83.5 (6.0)
	F1	23.5 (10.4)	48.6 (1.0)	37.5 (4.4)	95.5 (2.5)	<u>98.6 (0.1)</u>	83.3 (6.1)
LGCNFF	ACC	66.1 (1.0)	52.4 (2.1)	44.2 (0.6)	96.1 (0.3)	98.5 (0.2)	68.8 (0.4)
	F1	43.6 (4.4)	50.9 (2.1)	41.2 (1.0)	<u>96.1 (0.3)</u>	98.5 (0.2)	68.7 (0.4)
GEGCN	ACC	60.1 (1.8)	<u>58.6 (1.0)</u>	42.9 (0.4)	94.7 (0.1)	92.8 (4.7)	69.6 (0.4)
	F1	40.7 (4.2)	<u>58.6 (0.9)</u>	42.8 (0.3)	94.9 (0.3)	90.3 (6.5)	69.5 (0.4)
Ours	ACC	72.5 (1.5)	62.2 (0.7)	49.7 (0.6)	97.1 (0.2)	99.2 (0.1)	87.6 (0.1)
	F1	45.6 (5.3)	62.1 (0.9)	48.5 (0.6)	97.0 (0.1)	99.2 (0.1)	87.4 (0.1)

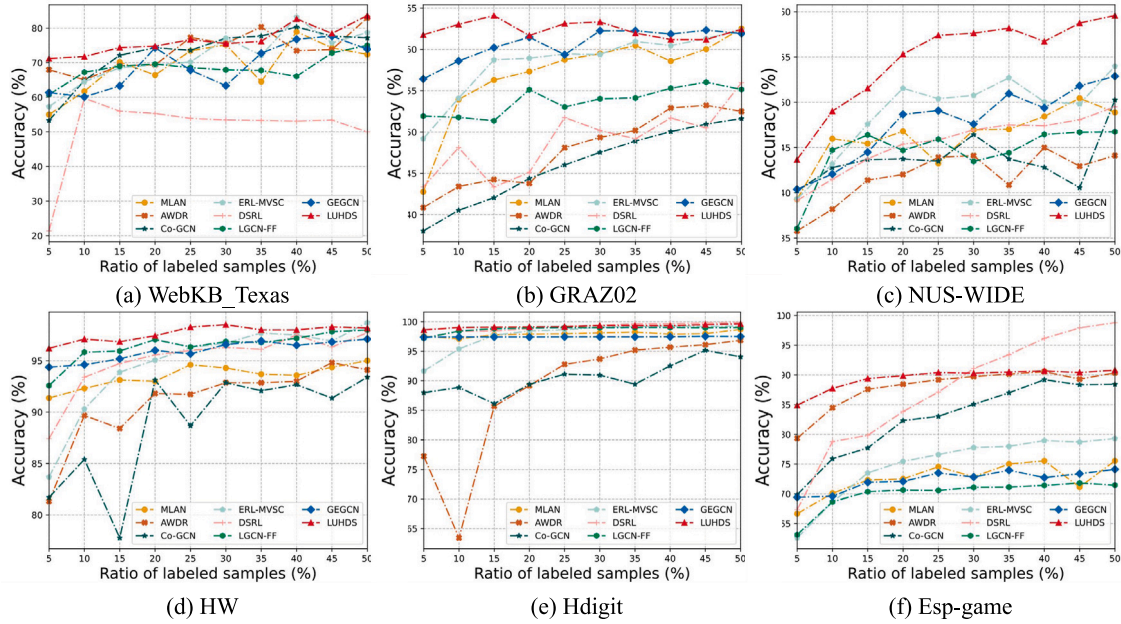


Fig. 3. The performance (Accuracy %) of all compared methods on tested datasets with the ratio of labeled samples ranging from 5% to 50%.

Performance comparison. We evaluate the performance of the proposed method and other compared algorithms on all tested datasets with widely used metrics of accuracy (ACC) and F1-score. Higher values of metrics indicate better performance. All methods are run five times and their average results along with standard deviations are presented in Table 3.

From the obtained experimental results, we have the following observations:

- The proposed method achieves the highest metrics of accuracy and F1-score on all datasets, which indicates the superiority of the proposed method in terms of the semi-supervised classification task.
- The proposed method exhibits remarkable improvements on WebKB_Texas and NUS-WIDE datasets, showcasing its capability to effectively capture higher-order relationships among views of small datasets.

Additionally, we conduct experiments on the compared methods on tested datasets with the ratio of labeled data ranging from 5% to 50%,

as shown in Fig. 3. It can be inferred from the figure that the proposed method is particularly well suited for the multi-view semi-supervised classification task, as it shows impressive performance even when the sample labeling rate is limited. On the contrary, other compared methods often require a large amount of supervisory information to show competitive performance. Overall, the proposed method demonstrates its superiority compared to other state-of-art algorithms.

4.4. Parameter sensitivity analysis

In this subsection, we investigate the sensitivity of the hyper-parameter T which is involved in the proposed method. We test the performance of the proposed method for hyper-parameter T in the range of $\{1, 2, 4, 8, 16, 32, 64\}$. We record the performance with ACC and plot a line graph of the model performance as a function of the parameter T as shown in Fig. 4.

Upon analyzing the figure, we can observe that the performance of the proposed method on most datasets shows an increasing and then a smooth trend with the parameter T . Specifically, in most cases, the performance of the proposed method rises with T and reaches its

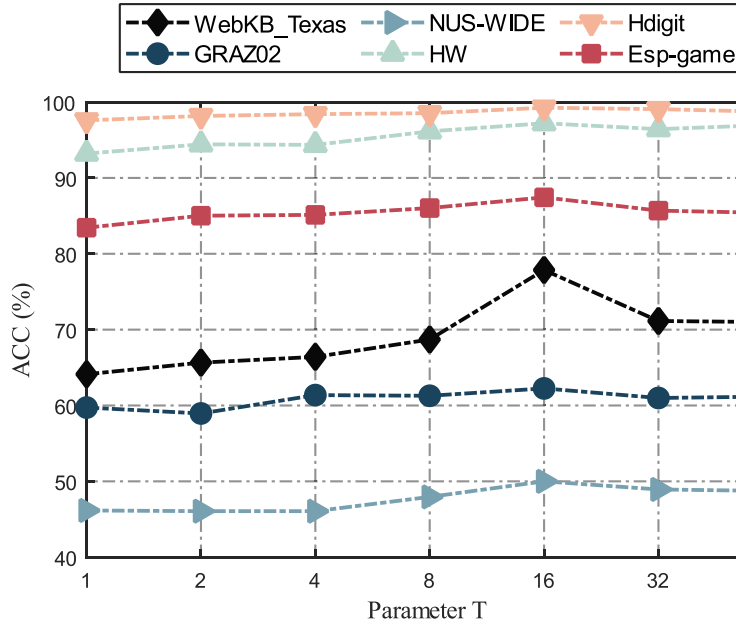


Fig. 4. The performance (Accuracy %) of the proposed method on all tested datasets with the number of T ranging from 1 to 64.

Table 4
Ablation study of the proposed method.

Datasets\Methods	Base-1	Base-2	LUHDS
WebKB_Texas	69.47	68.70	72.50
GRAZ02	59.54	61.96	63.20
NUS-WIDE	49.11	49.11	50.45
HW	93.29	96.93	97.21
Hdigit	98.41	99.13	99.21
Esp-game	86.83	86.29	87.70

maximum when T reaches 16, and although the performance decreases when T exceeds 16, still maintains a very competitive performance. The performance of the proposed method fluctuates significantly on the WebKB_Texas dataset, which may be due to the small number of sample points in WebKB_Texas dataset.

4.5. Ablation study

In this subsection, we conduct ablation experiments to evaluate the effectiveness of different modules in our model. We design the following two base modules:

- Base-1: It uses view-specific hypergraphs to train then fuses the outputs as the final node representation, which implies that the higher-order relationship among views is not utilized in training.
- Base-2: It removes the learnable unified cross-view hypergraph, which implies that the unified cross-view hypergraph is static during the training process.

We provide the results of the ablation experiments in Table 4. It can be observed from the table that the proposed method achieves the highest performance, which indicates the modules in our model is effective. Notably, the performance of Base-1 has a noticeable degradation compared to Base-2 on HW dataset, suggesting that the higher-order relationship among views on HW dataset is of importance for the semi-supervised classification task. This phenomenon reinforces the effectiveness of the unified cross-view hypergraph which can capture the higher-order relationship among views.

Moreover, in order to further illustrate the effectiveness of unified cross-view hypergraph, We compare U_1 , U_2 , U_3 , U_4 , and U_5 with the

second-best result in Table 3 on different datasets, and the results are shown in Fig. 5. It is evident that the learnable unified cross-view hypergraph generation strategy (U_5) achieves the highest performance. It should be highlighted that all unified cross-view hypergraphs generated based on different strategies outperform the second-best result in general, which also reflects the effectiveness and superiority of unified cross-view hypergraph. Although the performance of K-Means strategy and joint entropy strategy are lower than second-best algorithm on Hdigit dataset, the difference in their performance is not significant. The gap may be caused by the fact that the clustering algorithm and joint entropy are not suitable for this dataset. The implementation details of different hypergraph generation strategies are as follows:

- For U_1 that unified cross-view hypergraph constructed by k NN, we fix k as 15.
- For U_2 that unified cross-view hypergraph constructed by K -Means, we fix K as 8.
- For U_4 that unified cross-view hypergraph constructed by joint entropy, we set the γ as 0.1.

4.6. Visualization

In this subsection, we present visualizations of the learned representations of both our method and the compared algorithms.

Here we empirically employ t -SNE to visually represent the learned node representations of the proposed method and other algorithms on HW dataset, as shown in Fig. 6. It is evident that the proposed method gives a clearer representation that the different clusters are partitioned more explicitly, which suggests that the learned representation is discriminative.

4.7. Convergence analysis

In this subsection, we examine the convergence of the proposed method by analyzing the trends of evaluation metrics and loss values on different datasets, as exhibited in Fig. 7.

The results show that the loss values plunge quickly and eventually converge within a certain number of epochs on all tested datasets. The accuracy and F1-score increase rapidly and eventually fluctuate within a certain range with the epoch number growth during the

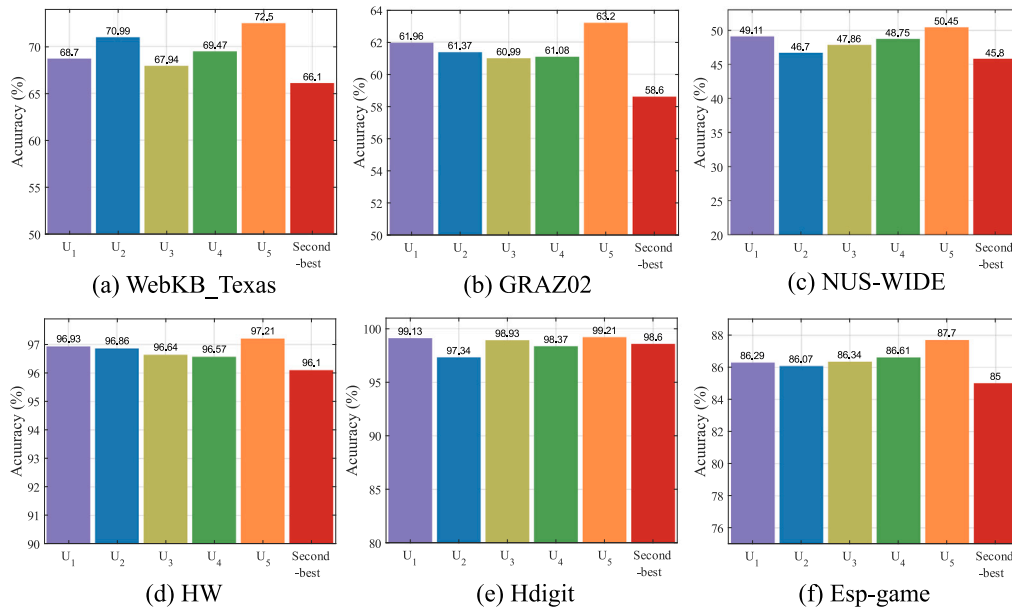


Fig. 5. Comparison of different hypergraph generation strategies and second-best result.

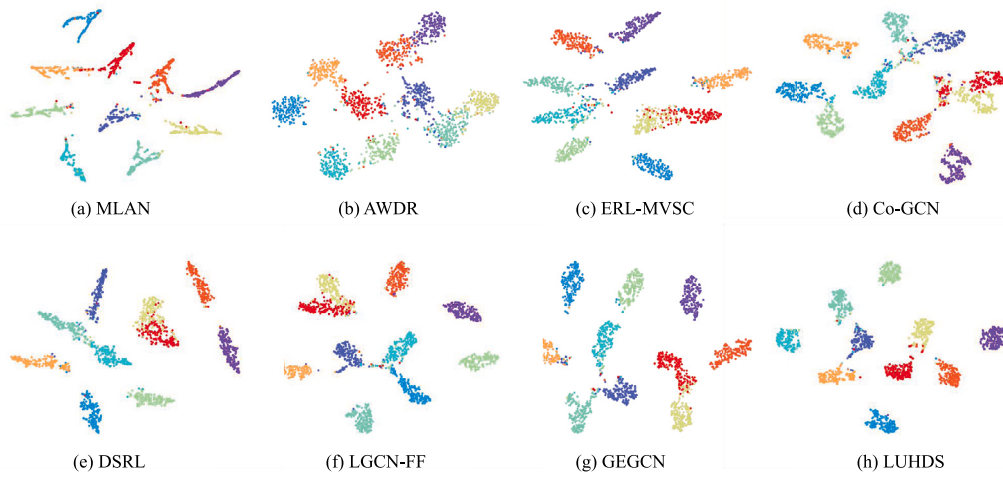


Fig. 6. Visualization results of the representations of compared algorithms on HW dataset.

training procedure. It is worth mentioning that the proposed method demonstrates swift convergence, often reaching convergence within 100 epochs on most datasets. Consequently, we can conclude that the proposed model exhibits a good convergence.

5. Conclusion

In this paper, we designed an end-to-end model Learnable unified hypergraph dynamic system for multi-view semi-supervised classification. Four strategies of unified cross-view hypergraph generation and a mechanism of learnable unified cross-view hypergraph generation were introduced in this paper. Furthermore, we utilized a dynamic diffusion system to perform multi-view semi-supervised classification and jointly optimize the hypergraph structure to make it suitable for the classification. Extensive quantitative experiments demonstrated the efficacy of the proposed method, with consistent improvements over the state-of-the-art baselines.

While the proposed methodology demonstrates promising results, it also encounters several limitations. First, the computational complexity

of dynamic diffusion processes and control steps may pose a challenge for high-dimension datasets. Second, the control function used in this model was implemented with only a simple linear layer, which perhaps limits its expressive power. These limitations highlight areas where further refinement is needed. Going forward, there are several avenues to further refine and improve the proposed method. First, the features of multi-view data are rich in information, and how to perform effective feature extraction by deep neural networks is yet to be explored. In addition, it is worthwhile to further investigate how to design a more efficient control function. Future research will also focus on developing a more efficient diffusion matrix for hypergraph dynamic systems.

CRediT authorship contribution statement

Zhibin Shi: Writing – original draft, Software, Methodology. **Zhenghong Lin:** Writing – review & editing, Formal analysis, Data curation. **Weihong Lin:** Visualization. **Shiping Wang:** Supervision.

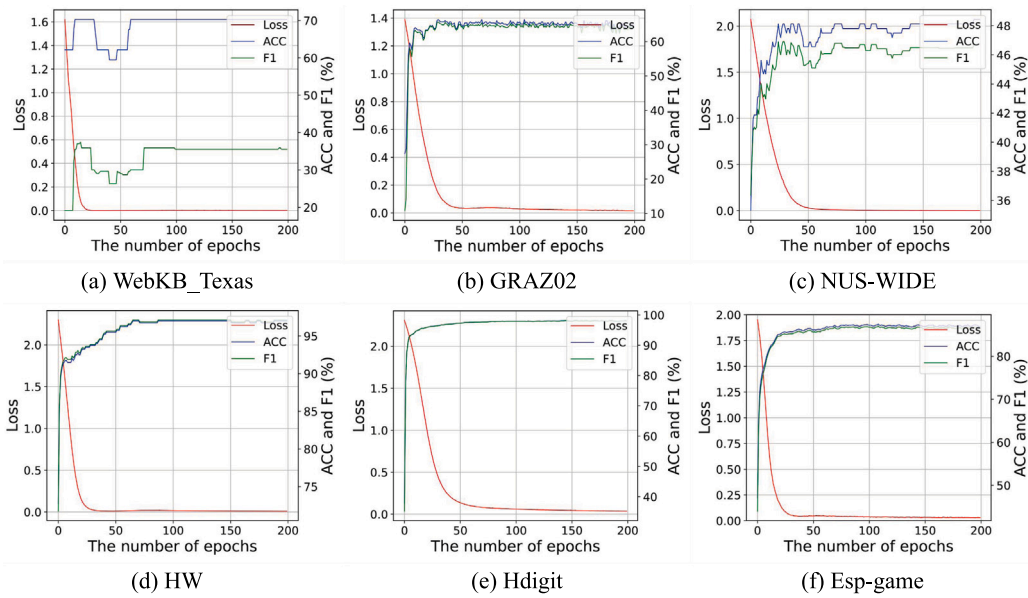


Fig. 7. Curves of loss values (red), accuracy (blue) and F1-score (green) of the proposed method.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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